National University of Computer and Emerging Sciences Islamabad
Operating System Lab Spring 2025

Lab Task: Named Pipes & dup &dup 2

Task 1: Chat Application Using Named Pipes

Objective:
Create a simple chat system between two processes using named pipes, where each process can send and receive messages asynchronously.

Steps:

1. Create two named pipes (`chat_in` and `chat_out`) using `mkfifo()`.
2. Implement **Process A**:
    - Reads user input and writes it to `chat_in`.
    - Simultaneously listens for messages from `chat_out`.
3. Implement **Process B**:
    - Reads messages from `chat_in` and responds via `chat_out`.
4. Allow continuous conversation until one of the processes sends "bye".
5. Clean up pipes using `unlink()` after termination.

Expected Input/Output:
Process A:

```
You: Hey, how are you?
Friend: I'm good! How about you?
You: bye
```

Process B:

```
Friend: Hey, how are you?
You: I'm good! How about you?
Friend: bye
Cleaning up chat pipes...
```

Task 2: Logging System Using dup() and dup2()

Objective:
Redirect standard output to a log file using `dup()` and `dup2()`, ensuring all program logs are stored in a file while still displaying critical messages on the console.

Steps:

1. Open a log file (`logfile.txt`) using `open()`.
2. Use `dup()` to duplicate the file descriptor.
3. Use `dup2()` to redirect standard output (`stdout`) to the log file.
4. Print general logs (e.g., "Program started", "Processing data") that go into the file.
5. Use the original file descriptor to print critical alerts to the console.
6. Close all file descriptors at the end.

Expected Input/Output:

Console Output:

```
Critical Alert: Disk space low!
```

Contents of logfile.txt:

```
Program started
Processing data...
Operation successful
```

Final Notes:

- Ensure the **reader process starts first** before the writer.
- Use **non-blocking reads** where needed to prevent deadlocks.

Instructions for Submission

- Ensure your code is **well-documented with comments**.
- Implement **proper error handling** (e.g., file not found, read/write failure, pipe issues).

- Test your program with **various inputs** before submission.
- Submit your **source code along with screenshots of successful runs**.

THE END
